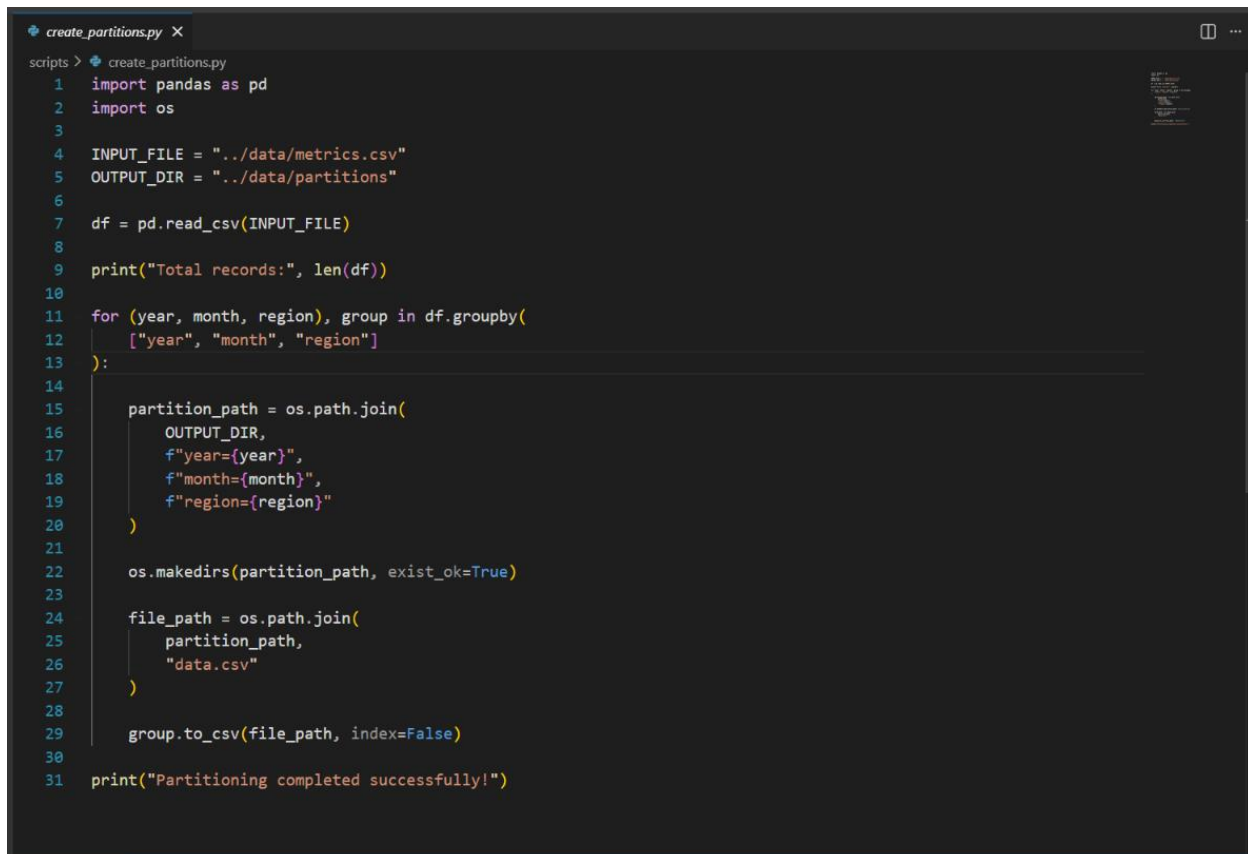


PRACTICAL -5

Aim:

To implement data storage structures using distributed storage paradigms by partitioning a large analytical dataset using Hive-style partitioning based on year and region, storing the partitioned data in MinIO object storage using Python and Boto3, querying specific partitions, and comparing the performance with a full-table scan.

Code Screenshot:



```
create_partitions.py X
scripts > create_partitions.py
1 import pandas as pd
2 import os
3
4 INPUT_FILE = "../data/metrics.csv"
5 OUTPUT_DIR = "../data/partitions"
6
7 df = pd.read_csv(INPUT_FILE)
8
9 print("Total records:", len(df))
10
11 for (year, month, region), group in df.groupby(
12     ["year", "month", "region"]
13 ):
14
15     partition_path = os.path.join(
16         OUTPUT_DIR,
17         f"year={year}",
18         f"month={month}",
19         f"region={region}"
20     )
21
22     os.makedirs(partition_path, exist_ok=True)
23
24     file_path = os.path.join(
25         partition_path,
26         "data.csv"
27     )
28
29     group.to_csv(file_path, index=False)
30
31 print("Partitioning completed successfully!")
```

```
generate_data.py X
scripts > generate_data.py > ...
1 import pandas as pd
2 import random
3 from datetime import datetime, timedelta
4
5 NUM_RECORDS = 20000
6
7 regions = [
8     "Gujarat",
9     "Maharashtra",
10    "Rajasthan",
11    "Delhi",
12    "Karnataka"
13 ]
14
15 applications = [
16     "MobileApp",
17     "WebApp",
18     "API",
19     "Payment",
20     "Analytics"
21 ]
22
23 metrics = [
24     "response_time",
25     "cpu_usage",
26     "memory_usage",
27     "request_count",
28     "error_count"
29 ]
30
31 start_date = datetime(2022, 1, 1)
32
33 data = []
34
```

```
generate_data.py X
scripts > generate_data.py > ...
33 data = []
34
35 for i in range(NUM_RECORDS):
36
37     timestamp = start_date + timedelta(
38         minutes=random.randint(0, 4 * 365 * 24 * 60)
39     )
40
41     data.append({
42         "timestamp": timestamp,
43         "year": timestamp.year,
44         "month": timestamp.month,
45         "region": random.choice(regions),
46         "application": random.choice(applications),
47         "metric": random.choice(metrics),
48         "value": round(random.uniform(10, 1000), 2)
49     })
50
51 df = pd.DataFrame(data)
52
53 df.to_csv("../data/metrics.csv", index=False)
54
55 print("Dataset generated successfully!")
56 print("Records:", len(df))
57 print()
58 print(df.head())
```

```
minio_query.py X
scripts > minio_query.py > ...
1  import boto3
2  import pandas as pd
3  import io
4  import time
5
6  s3 = boto3.client(
7      "s3",
8      endpoint_url="http://localhost:9000",
9      aws_access_key_id="admin",
10     aws_secret_access_key="Minio@12345"
11 )
12
13 BUCKET = "metrics"
14
15 YEAR = 2025
16 REGION = "Gujarat"
17
18 prefix = f"year={YEAR}/"
19
20 start_time = time.perf_counter()
21
22 response = s3.list_objects_v2(
23     Bucket=BUCKET,
24     Prefix=prefix
25 )
26
27 matching_files = []
28
29 for obj in response.get("Contents", []):
30
31     key = obj["Key"]
32
33     if (
34         f"region={REGION}" in key
```

```
partition_query.py X
scripts > partition_query.py > ...
1 import pandas as pd
2 import glob
3 import time
4
5 YEAR = 2025
6 REGION = "Gujarat"
7
8 pattern = (
9     f"../data/partitions/"
10    f"year={YEAR}/"
11    f"month=*/"
12    f"region={REGION}/"
13    f"data.csv"
14 )
15
16 files = glob.glob(pattern)
17
18 print("Partitions found:", len(files))
19
20 start_time = time.perf_counter()
21
22 dataframes = []
23
24 for file in files:
25     df = pd.read_csv(file)
26     dataframes.append(df)
27
28 result = pd.concat(dataframes, ignore_index=True)
29
30 end_time = time.perf_counter()
31
32 elapsed = end_time - start_time
33
34 print("Records found:", len(result))
```

Output Screenshot:

```
Administrator: Command Prompt

C:\Windows\System32>docker --version
Docker version 29.7.2, build a7dca66

C:\Windows\System32>docker info
Client:
 Version:      29.7.2
 Context:      desktop-linux
 Debug Mode:   false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.119.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.30.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.36.0-desktop.1
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.3.1
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.3
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-desktop.exe
  dhis: cli for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.7
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.43.3
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.2.6
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.9
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-offload.exe
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.2.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-pass.exe
  sandbox: "docker sandbox" is deprecated, use Docker Sandboxes instead (Docker Inc.)
    Version:  v0.13.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-sandbox.exe

Administrator: Command Prompt

    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.9
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-offload.exe
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.2.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-pass.exe
  sandbox: "docker sandbox" is deprecated, use Docker Sandboxes instead (Docker Inc.)
    Version:  v0.13.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-sandbox.exe
  scout: Docker Scout (Docker Inc.)
    Version:  v1.24.0
    Path:      C:\Users\Digisha\.docker\cli-plugins\docker-scout.exe

Server:
 Containers: 1
  Running: 1
  Paused: 0
  Stopped: 0
 Images: 1
 Server Version: 29.7.2
 Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Discovered Devices:
  cdi: docker.com/gpu-webgpu
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 nvidia runc
 Default Runtime: runc
 Init Binary: docker-init
  containerd version: e337c1516c3b2bffa98eb76f1f4117477e6f4e66
  runc version: v1.3.6-0-g491b69ba
  init version: de4bad0
 Security Options:
  seccomp
   Profiles: builtin
  cgroupns
 Kernel Version: 6.18.33.2-microsoft-standard-WSL2
 Operating System: Docker Desktop
 OSType: linux
 Architecture: x86_64
 CPUs: 12
 Total Memory: 7.581GiB
```

```
C:\Windows\System32>docker run -d --name minio -p 9000:9000 -p 9001:9001 -e MINIO_ROOT_USER=admin -e MINIO_ROOT_PASSWORD=Minio@12345 minio/minio server /data --console-address ":9001"

What's next:
  Debug this container error with Gordon + docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/minio" is already in use by container "b3e2e262b460bfa5bfc8518422e845cc932468f242740f8ebce5e8e18b3". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

C:\Windows\System32>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
C:\Windows\System32>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
b3e2e262b460   minio/minio  "/usr/bin/docker-ent..."  20 hours ago   Exited (255) 4 minutes ago   0.0.0.0:9000-9001->9000-9001/tcp   minio

C:\Windows\System32>docker start minio
minio

C:\Windows\System32>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
b3e2e262b460   minio/minio  "/usr/bin/docker-ent..."  20 hours ago   Up 8 seconds   0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp   minio

C:\Windows\System32>
```



```
Windows PowerShell
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PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python --version
Python 3.14.3
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> pip --version
pip 26.1.1 from C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages\pip (python 3.14)
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> pip install boto3 pandas
Requirement already satisfied: boto3 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (1.43.69)
Requirement already satisfied: pandas in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (3.0.2)
Requirement already satisfied: botocore<1.44.0,>=1.43.69 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from boto3) (1.43.69)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from boto3) (1.1.0)
Requirement already satisfied: s3transfer<0.20.0,>=0.19.0 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from boto3) (0.19.2)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from botocore<1.44.0,>=1.43.69->boto3) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from botocore<1.44.0,>=1.43.69->boto3) (2.6.3)
Requirement already satisfied: six>=1.5 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.44.0,>=1.43.69->boto3) (1.17.0)
Requirement already satisfied: numpy>=2.3.3 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2.4.4)
Requirement already satisfied: tzdata in C:\Users\Digisha\AppData\Roaming\Python\Python314\site-packages (from pandas) (2025.3)

[notice] A new release of pip is available: 26.1.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python -c "import boto3; import pandas; print('Boto3 and Pandas OK!')"
Boto3 and Pandas OK
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> ls

Directory: D:\sem 5\data engineering\practical\p5\minio_partition_lab

Mode                LastWriteTime         Length Name
----                -
d-----          12-08-2026         10:58      data
d-----          12-08-2026         10:58    scripts
-a-----          10-08-2026         15:55      332 test_minio.py

PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python generate_data.py
Dataset generated successfully!
```



```

Windows PowerShell
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python generate_data.py
Dataset generated successfully!
Records: 20000

timestamp year month region application metric value
0 2022-01-02 11:08:00 2022 1 Karnataka Analytics request_count 458.63
1 2025-09-01 07:04:00 2025 9 Rajasthan MobileApp response_time 636.49
2 2023-09-30 01:19:00 2023 9 Gujarat Payment request_count 145.37
3 2024-04-21 07:48:00 2024 4 Maharashtra MobileApp cpu_usage 913.09
4 2025-03-18 05:20:00 2025 3 Gujarat Analytics cpu_usage 596.43
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> cd ..
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python -c "import pandas as pd; df=pd.read_csv('data/metrics.csv'); print(df.shape); print(df.head())"
(20000, 7)

timestamp year month region application metric value
0 2022-01-02 11:08:00 2022 1 Karnataka Analytics request_count 458.63
1 2025-09-01 07:04:00 2025 9 Rajasthan MobileApp response_time 636.49
2 2023-09-30 01:19:00 2023 9 Gujarat Payment request_count 145.37
3 2024-04-21 07:48:00 2024 4 Maharashtra MobileApp cpu_usage 913.09
4 2025-03-18 05:20:00 2025 3 Gujarat Analytics cpu_usage 596.43
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python partition_data.py
C:\Users\Digisha\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts\partition_data.py': [Errno 2] No such file or directory
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> cd ..
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python test_minio.py
Connected to MinIO successfully!
Buckets:
- metrics
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> tree data /f
Folder PATH listing for volume collage
Volume serial number is 000001EC FE20:D825
D:\SEM 5\DATA ENGINEERING\PRACTICAL\P5\MINIO_PARTITION_LAB\DATA
├── metrics.csv
├── partitions
│   ├── year=2022
│   │   ├── month=1
│   │   │   ├── region=Delhi
│   │   │   └── data.csv

```

```
Windows PowerShell
Folder PATH listing for volume collage
Volume serial number is 886881EC F228:2B25
D:\SEM 5\DATA ENGINEERING\PRACTICAL\PSYMINIO_PARTITION_LAB\DATA
metrics.csv

partitions
├── year=2022
│   ├── month=1
│   │   ├── region=Delhi
│   │   │   └── data.csv
│   │   ├── region=Gujarat
│   │   │   └── data.csv
│   │   ├── region=Karnataka
│   │   │   └── data.csv
│   │   ├── region=Maharashtra
│   │   │   └── data.csv
│   │   └── region=Rajasthan
│   │       └── data.csv
│   ├── month=10
│   │   ├── region=Delhi
│   │   │   └── data.csv
│   │   ├── region=Gujarat
│   │   │   └── data.csv
│   │   ├── region=Karnataka
│   │   │   └── data.csv
│   │   ├── region=Maharashtra
│   │   │   └── data.csv
│   │   └── region=Rajasthan
│   │       └── data.csv
│   ├── month=11
│   │   ├── region=Delhi
│   │   │   └── data.csv
│   │   ├── region=Gujarat
│   │   │   └── data.csv
│   │   ├── region=Karnataka
│   │   │   └── data.csv
│   │   ├── region=Maharashtra
│   │   │   └── data.csv
│   │   └── region=Rajasthan
│   │       └── data.csv
│   └── month=12
│       ├── region=Delhi
│       │   └── data.csv
│       ├── region=Gujarat
│       │   └── data.csv
│       ├── region=Karnataka
│       │   └── data.csv
│       └── region=Maharashtra
│           └── data.csv
└── year=2023
    ├── month=1
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=2
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=3
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=4
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=5
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=6
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=7
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=8
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=9
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=10
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    ├── month=11
    │   ├── region=Delhi
    │   │   └── data.csv
    │   ├── region=Gujarat
    │   │   └── data.csv
    │   ├── region=Karnataka
    │   │   └── data.csv
    │   ├── region=Maharashtra
    │   │   └── data.csv
    │   └── region=Rajasthan
    │       └── data.csv
    └── month=12
        ├── region=Delhi
        │   └── data.csv
        ├── region=Gujarat
        │   └── data.csv
        ├── region=Karnataka
        │   └── data.csv
        ├── region=Maharashtra
        │   └── data.csv
        └── region=Rajasthan
            └── data.csv
```



```

Windows PowerShell
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python create_partitions.py
Total records: 20000
Partitioning completed successfully!
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python upload_partitions.py
Uploading: year=2022/month=1/region=Delhi/data.csv
Uploading: year=2022/month=1/region=Gujarat/data.csv
Uploading: year=2022/month=1/region=Karnataka/data.csv
Uploading: year=2022/month=1/region=Maharashtra/data.csv
Uploading: year=2022/month=1/region=Rajasthan/data.csv
Uploading: year=2022/month=10/region=Delhi/data.csv
Uploading: year=2022/month=10/region=Gujarat/data.csv
Uploading: year=2022/month=10/region=Karnataka/data.csv
Uploading: year=2022/month=10/region=Maharashtra/data.csv
Uploading: year=2022/month=10/region=Rajasthan/data.csv
Uploading: year=2022/month=11/region=Delhi/data.csv
Uploading: year=2022/month=11/region=Gujarat/data.csv
Uploading: year=2022/month=11/region=Karnataka/data.csv
Uploading: year=2022/month=11/region=Maharashtra/data.csv
Uploading: year=2022/month=11/region=Rajasthan/data.csv
Uploading: year=2022/month=12/region=Delhi/data.csv
Uploading: year=2022/month=12/region=Gujarat/data.csv
Uploading: year=2022/month=12/region=Karnataka/data.csv
Uploading: year=2022/month=12/region=Maharashtra/data.csv
Uploading: year=2022/month=12/region=Rajasthan/data.csv
Uploading: year=2022/month=2/region=Delhi/data.csv
Uploading: year=2022/month=2/region=Gujarat/data.csv
Uploading: year=2022/month=2/region=Karnataka/data.csv
Uploading: year=2022/month=2/region=Maharashtra/data.csv
Uploading: year=2022/month=2/region=Rajasthan/data.csv
Uploading: year=2022/month=3/region=Delhi/data.csv
Uploading: year=2022/month=3/region=Gujarat/data.csv
Uploading: year=2022/month=3/region=Karnataka/data.csv
Uploading: year=2022/month=3/region=Maharashtra/data.csv
Uploading: year=2022/month=3/region=Rajasthan/data.csv
Uploading: year=2022/month=4/region=Delhi/data.csv
Uploading: year=2022/month=4/region=Gujarat/data.csv
Uploading: year=2022/month=4/region=Karnataka/data.csv
Uploading: year=2022/month=4/region=Maharashtra/data.csv
Uploading: year=2022/month=4/region=Rajasthan/data.csv
Uploading: year=2022/month=5/region=Delhi/data.csv
Uploading: year=2022/month=5/region=Gujarat/data.csv
Uploading: year=2022/month=5/region=Karnataka/data.csv

Uploading: year=2025/month=7/region=Delhi/data.csv
Uploading: year=2025/month=7/region=Gujarat/data.csv
Uploading: year=2025/month=7/region=Karnataka/data.csv
Uploading: year=2025/month=7/region=Maharashtra/data.csv
Uploading: year=2025/month=7/region=Rajasthan/data.csv
Uploading: year=2025/month=8/region=Delhi/data.csv
Uploading: year=2025/month=8/region=Gujarat/data.csv
Uploading: year=2025/month=8/region=Karnataka/data.csv
Uploading: year=2025/month=8/region=Maharashtra/data.csv
Uploading: year=2025/month=8/region=Rajasthan/data.csv
Uploading: year=2025/month=9/region=Delhi/data.csv
Uploading: year=2025/month=9/region=Gujarat/data.csv
Uploading: year=2025/month=9/region=Karnataka/data.csv
Uploading: year=2025/month=9/region=Maharashtra/data.csv
Uploading: year=2025/month=9/region=Rajasthan/data.csv

All partitions uploaded successfully!
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python partition_query.py
Partitions found: 12
Records found: 1000
Partitions scanned: 12
Time: 59.45 ms
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python performance_test.py

===== FULL TABLE SCAN =====
Records found: 1000
Records scanned: 20000
Time: 112.779 ms

===== PARTITIONED SCAN =====
Records found: 81
Records scanned: 81
Time: 4.375 ms

===== PERFORMANCE COMPARISON =====
Full table scan: 112.779 ms
Partition scan: 4.375 ms
Performance improvement: 96.12 %
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> |

```

The image displays two software interfaces. The top interface is the MinIO Object Store 'Object Browser'. It shows a bucket named 'metrics' with four sub-directories: 'year-2022', 'year-2023', 'year-2024', and 'year-2025'. The bucket was created on Wed, Aug 12 2026 14:02:58 (GMT+5:30) and is set to PRIVATE access. The bottom interface is Docker Desktop, showing a container named 'minio' running. The container's status is 'Running (22 minutes ago)'. The logs for the container show the MinIO Object Storage Server startup information, including the version (RELEASE.2025-09-07T16-13-09Z) and the license (GNU AGPLv3).

MinIO Object Store Object Browser

Created on: Wed, Aug 12 2026 14:02:58 (GMT+5:30) Access: PRIVATE

Name	Last Modified	Size
year-2022	-	-
year-2023	-	-
year-2024	-	-
year-2025	-	-

Docker Desktop

Containers / minio

minio
 b3e2e262b460
 9000:9000
 minio/minio:latest

STATUS: Running (22 minutes ago)

Logs

```
INFO: Formatting 1st pool, 1 set(s), 1 drives per set.
INFO: WARNING: Host local has more than 0 drives of set. A host failure will result in data becoming unavailable.
MinIO Object Storage Server
Copyright: 2015-2026 MinIO, Inc.
License: GNU AGPLv3 - https://www.gnu.org/licenses/agpl-3.0.html
Version: RELEASE.2025-09-07T16-13-09Z (go1.24.6 linux/amd64)

API: http://172.17.0.2:9000 http://127.0.0.1:9000
WebUI: http://172.17.0.2:9001 http://127.0.0.1:9001

Docs: https://docs.min.io
MinIO Object Storage Server
Copyright: 2015-2026 MinIO, Inc.
License: GNU AGPLv3 - https://www.gnu.org/licenses/agpl-3.0.html
Version: RELEASE.2025-09-07T16-13-09Z (go1.24.6 linux/amd64)

API: http://172.17.0.2:9000 http://127.0.0.1:9000
WebUI: http://172.17.0.2:9001 http://127.0.0.1:9001

Docs: https://docs.min.io
```

Engine running | RAM 1.14 GB CPU 0.42% Disk: 1.68 GB used (limit 1006.85 GB) | Terminal v4.86.0

Learning Outcome:

After completing this practical, I learned how to organize analytical data using Hive-style partitioning and store it in an S3-compatible MinIO object storage system. I learned to use Python and Boto3 to upload and retrieve partitioned data, query only the required partitions, and compare partition-based access with full-table scanning. I also understood how partitioning can reduce unnecessary data access and improve the efficiency of analytical queries.